

BENCHMARK: NON-DETECTION

Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2457824.71008 +/- 0.00037 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.091 +/- 0.044                   |
| Transit depth (Rp/Rs)^2                           | 0.82 +/- 0.8 [%]                  |
| Orbital Inclination (inc)                         | 76.88 +/- 2.82                    |
| Airmass coefficient 1 (a1)                        | 0.47 +/- 0.31                     |
| Airmass coefficient 2 (a2)                        | 0.52 +/- 0.24                     |
| Scatter in the residuals of the lightcurve fit is | 67.14 %                           |
| Best Comparison Star                              | None                              |
| Optimal Aperture                                  | 3.0                               |
| Optimal Annulus                                   | 3.5                               |
| Transit Duration (day)                            | 0.098 +/- 0.017                   |

Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.091 ± 0.044 vs 0.1178 (deviation 0.61σ) |
| Duration fit vs published | 0.098 d vs 0.125 d (deviation 1.59σ)      |
| Mid-time O–C              | 1571.2 min                                |
| Depth SNR                 | 2.1                                       |
| Residual scatter          | 67.14 %                                   |

Light curve

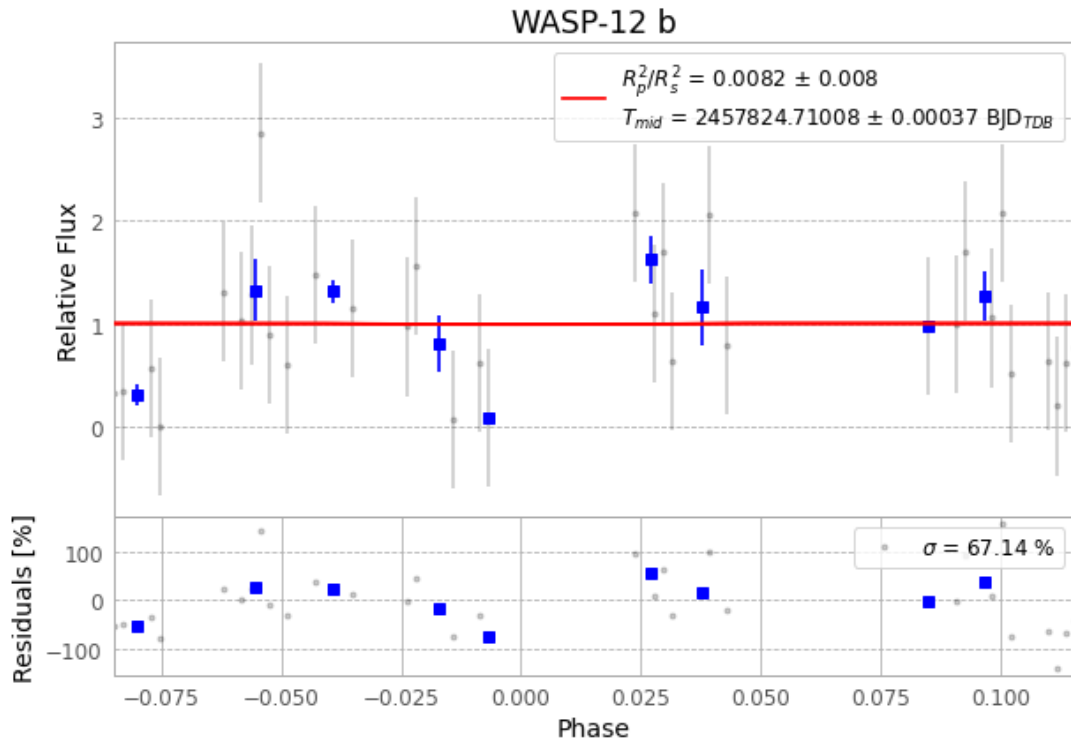


Figure 1 — FinalLightCurve\_WASP-12 b\_2017-03-11.png (SHA-256 0d77719eea7ccec7...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [279, 358], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 dd64ec00e5afa151...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit b365494

## Data provenance

107 FITS frames (70 MB), WASP-12170312024212.FITS → WASP-12170312080012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-170311.log (4bd8acad250d...), FinalLightCurve\_WASP-12 b\_2017-03-11.png (0d77719eea7c...), FinalLightCurve\_WASP-12 b\_2017-03-11.csv (56de48790510...)

## Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-23 16:43Z.